FOR A FLAT MORPHISM THE SHRIEK OF THE STRUCTURE SHEAF IS A RELATIVE DUALIZING COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0E9W (sources/stacks-duality/pinned-original-source.tex line 7589).

Allowed standard prerequisites (editorial).

- Relative perfectness of the shriek of the structure sheaf for a flat morphism. Same chapter, Lemma labelled lemma-flat-shriek-relatively-perfect, cited at proof lines 7622-7624. Gives the first component of the relative dualizing structure.
- The description of the diagonal pushforward through the sheaf with exact support. Same chapter, Lemmas labelled lemma-twisted-inverse-image-closed and lemma-sheaf-with-exact-support-ext, cited at proof lines 7626-7634. Turns the condition into a computation of Delta^!.
- Base change of the shriek along a flat morphism. Same chapter, Lemma labelled lemma-base-change-shriek-flat, cited at proof lines 7648-7652. Gives pr_1^! o Lpr_2^* = Lpr_1^* o f^!.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 0E2T: sources/stacks-duality/pinned-original-source.tex, source lines 7234–7234. Same-chapter pointer into the author environment definition at source line 7234; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0E9T: sources/stacks-duality/pinned-original-source.tex, source lines 4890–4890. Same-chapter pointer into the author environment lemma at source line 4890; invoked at proof line 7598 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9X: sources/stacks-duality/pinned-original-source.tex, source lines 2467–2467. Same-chapter pointer into the author environment lemma at source line 2467; invoked at proof line 7605 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A77: sources/stacks-duality/pinned-original-source.tex, source lines 2339–2339. Same-chapter pointer into the author environment lemma at source line 2339; invoked at proof line 7606 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0E9U: sources/stacks-duality/pinned-original-source.tex, source lines 4997–4997. Same-chapter pointer into the author environment lemma at source line 4997; invoked at proof line 7619 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let $f : X \to Y$ be a morphism of FTS_S. If f is flat, then $f^!\mathcal{O}_Y$ is (the first component of) a relative dualizing complex for X over Y in the sense of Definition 0E2T.*

Proof. By Lemma 0E9T we have that $f^!\mathcal{O}_Y$ is Y-perfect. As f is separated the diagonal $\Delta : X \to X \times_Y X$ is a closed immersion and $\Delta_*\Delta^!(-) = R\mathcal{H}om_{\mathcal{O}_{X\times_Y X}}(\mathcal{O}_X, -)$, see Lemmas 0A9X and 0A77. Hence to finish the proof it suffices to show $\Delta^!(L\mathrm{pr}_1^* f^!(\mathcal{O}_Y)) \cong \mathcal{O}_X$ where $\mathrm{pr}_1 : X \times_Y X \to X$ is the first projection. We have

$$\mathcal{O}_X = \Delta^!\mathrm{pr}_1^!\mathcal{O}_X = \Delta^!\mathrm{pr}_1^! L\mathrm{pr}_2^*\mathcal{O}_Y = \Delta^!(L\mathrm{pr}_1^* f^!\mathcal{O}_Y)$$

where $\mathrm{pr}_2 : X \times_Y X \to X$ is the second projection and where we have used the base change isomorphism $\mathrm{pr}_1^! \circ L\mathrm{pr}_2^* = L\mathrm{pr}_1^* \circ f^!$ of Lemma 0E9U. □